

Summary

Software Engineer with an M.S. in Electrical Engineering, building frontend/backend platforms, AI services, and robotics software for real-time systems and system integration. Experienced with FastAPI, WebSocket, React / TypeScript, Next.js, Docker, Linux, C# / .NET, ROS 2, Isaac GROOT, VLA, and computer-vision workflows including PatchCore.

Focus Areas

70% → 90% cable-insertion success	3-4x manual data volume in the same time window	-2 operators manual staffing needs
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Experience

- AMTRAN TECHNOLOGY CO., LTD. | AI Robotics EngineerMay 2025 - Present | New Taipei, Taiwan
- Owned software delivery from requirements and system architecture through Docker deployment, testing, on-site integration, and maintenance for real-time AI and robotics systems.
 - Built a Next.js / React Flow / FastAPI / WebSocket workflow and data-recording platform for automated task execution and AI data workflows, reaching approximately 3-4x the manual data volume in the same time window and reducing manual staffing needs by two operators.
 - Developed a React / TypeScript / FastAPI / ROS 2 web control and monitoring platform with real-time feedback, video, system status, control commands, and Docker / Nginx deployment; improved cable-insertion success from 70% to 90% in a robotics integration task.
 - Integrated ROS 2 control and simulation, Isaac GROOT N1.7 data conversion, VLA policy inference, visual perception, AI vision, and PatchCore anomaly detection across software and physical-robot workflows.
- ISCOM | Software Engineer InternFeb 2022 - Jun 2022 | Kaohsiung, Taiwan
- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Backend API & Services	FastAPI, Flask, ASP.NET MVC, WebSocket, REST APIs, MS SQL, PostgreSQL / SQLite
Frontend & Workflow Platforms	React, TypeScript, Next.js, React Flow, JavaScript, jQuery
AI / ML & Computer Vision	PatchCore anomaly detection, PyQt, PyTorch, OpenCV, YOLO OBB, AprilTag, Isaac GROOT, VLA
Robotics & Real-Time Systems	ROS 2, Isaac Sim / Lab, robot control, policy inference, Digital Twin workflows
Deployment & Systems	Docker, Nginx, Linux, Git, Python, C++, C# / .NET, API and data-pipeline integration

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place